JUNIOR SCHOOL LEARNING HUB

GRADE 9 PRE-TECHNICAL STUDIES
END-OF-TERM 3 EXAMINATION

Name: _____	Adm. No.: _____	Date: _____	Class: _____
Time: 2 hours 30 minutes	Total: 100 marks	Term: 3	Learning Area: Pre-Technical Studies

INSTRUCTIONS TO LEARNERS

- Answer all questions.
- Read all instructions carefully.
- Use neat, labelled sketches where required.
- Observe safety precautions when describing tools, materials and workshop procedures.

SECTION A: SHORT ANSWER QUESTIONS (40 MARKS)

1. Define a technical drawing. (2 marks)

2. State two reasons why dimensions are included on a working drawing. (2 marks)

3. Name two instruments used for drawing accurate straight lines or angles. (2 marks)

4. Differentiate between first-angle and third-angle orthographic projection. (2 marks)

5. State two uses of a scale in technical drawing. (2 marks)

6. What is a simple machine? (2 marks)

7. Name two types of levers and give one example of each. (2 marks)

8. Define mechanical advantage. (2 marks)

9. State two ways of reducing friction in a moving machine. (2 marks)

10. Give two examples of materials commonly used in making simple products. (2 marks)

11. State two properties of a good construction material. (2 marks)

12. Differentiate between a thermoplastic and a thermosetting plastic. (2 marks)

13. Name two joining methods used in product construction. (2 marks)

14. State two purposes of finishing a product. (2 marks)

15. Give two examples of renewable energy sources. (2 marks)

16. State two advantages of using solar energy. (2 marks)

17. What is an electrical circuit? (2 marks)

18. Name two components of a basic electrical circuit. (2 marks)

19. State two safety precautions when using hand tools in a workshop. (2 marks)

20. Give two ways of maintaining tools and equipment. (2 marks)

SECTION B: STRUCTURED QUESTIONS (60 MARKS)

21. Technical drawing and design communication (12 marks)

(a) Explain four reasons why technical drawings are important when making a product. (4)

(b) Describe four steps followed when preparing a neat orthographic drawing of a simple object. (4)

(c) State four details that may be included in a title block. (4)

22. Machines and mechanical systems (12 marks)

(a) Describe the function of each of the following simple machines: lever, pulley, inclined plane and wheel-and-axle. (4)

(b) A machine has a load of 240 N and an effort of 60 N. Calculate its mechanical advantage. Show your working. (2)

(c) Explain three ways in which simple machines make work easier. (6)

23. Materials, joining and finishing (12 marks)

(a) Explain four factors to consider when selecting materials for a product. (4)

(b) Describe four joining methods that may be used when making products. (4)

(c) Explain two reasons for applying a suitable finish to a product. (4)

24. Energy and basic electrical systems (12 marks)

(a) Explain three benefits of renewable energy in homes and communities. (3)

(b) Draw and label a simple electrical circuit containing a cell, switch, connecting wires and a bulb. (4)

(c) State three safety precautions to observe when working with electricity. (3)

(d) Give two examples of situations in which solar energy can be used. (2)

25. Workshop practice, safety and maintenance (12 marks)

(a) Explain four workshop safety rules and why each is important. (4)

(b) Describe four steps to follow when selecting and using a hand tool for a task. (4)

(c) State four practices that help keep a workshop organised and tools in good condition. (4)

MARKING SCHEME

Grade 9 Pre-Technical Studies – End-of-Term 3 Examination

Total: 100 marks | Section A: 40 | Section B: 60

SECTION A: SHORT ANSWER QUESTIONS (40 MARKS)

1. A precise graphic representation of an object or structure showing its shape, size and construction details. (2)

2. Any two: communicate exact sizes; guide manufacture/assembly; reduce errors and waste; ensure parts fit. (1 each)

3. Any two: ruler/scale, set square, protractor, T-square, compass. (1 each)

4. First-angle places the object between observer and projection plane and positions views opposite their viewing direction; third-angle places the projection plane between observer and object and positions views on the same side as viewed. (1 each)

5. Any two: represent large objects on smaller paper; enlarge small details; maintain proportional relationships. (1 each)

6. A device that makes work easier by changing the size or direction of a force, or the distance over which it acts. (2)

7. Any two correct pairs, e.g. first-class—seesaw/scissors; second-class—wheelbarrow/nutcracker; third-class—tweezers/fishing rod. (1 each)

8. The ratio of load to effort in a machine. (2)

9. Any two: lubrication; use of bearings; smoothing contact surfaces; correct alignment. (1 each)

10. Any two suitable examples: wood, metal, plastic, glass, ceramics, textiles. (1 each)

11. Any two: strength, durability, suitable weight, workability, corrosion resistance, availability, affordability. (1 each)

12. Thermoplastics soften when heated and can be reshaped repeatedly; thermosetting plastics set permanently and cannot be remoulded by reheating. (1 each)

13. Any two: welding, soldering, riveting, screwing/bolting, nailing, gluing, dovetail or other suitable joints. (1 each)

14. Any two: protect from corrosion/moisture; improve appearance; improve durability; provide a smooth surface. (1 each)

15. Any two: solar, wind, hydro, geothermal, biomass. (1 each)

16. Any two: renewable; low operating emissions; reduces electricity costs; useful in off-grid areas; low running costs after installation. (1 each)

17. A complete path through which electric current can flow. (2)

18. Any two: cell/battery, switch, bulb/load, connecting wires, resistor. (1 each)

19. Any two: wear appropriate PPE; use the correct tool; inspect tools before use; keep hands dry; carry tools safely; secure workpieces. (1 each)

20. Any two: clean after use; dry and store properly; sharpen where appropriate; lubricate moving parts; inspect and repair damaged tools. (1 each)

SECTION B: STRUCTURED QUESTIONS (60 MARKS)

21. Technical drawing and design communication (12 marks)

(a) Any four explained, 1 mark each: communicates design; gives accurate dimensions; guides construction; shows shape and arrangement; supports quality control; reduces mistakes and material waste. (4)

(b) Any four steps, 1 mark each: study object; choose suitable scale and views; draw light construction lines; project views accurately; add visible details; darken final outlines; add dimensions and labels. (4)

(c) Any four: drawing title; learner/designer name; scale; date; drawing number; material; projection symbol; institution/class. (4)

22. Machines and mechanical systems (12 marks)

(a) Lever: rigid bar turning about a fulcrum; pulley: grooved wheel with rope/cord to change force direction or magnitude; inclined plane: sloping surface that reduces effort over a longer distance; wheel-and-axle: large wheel fixed to a smaller axle to transmit/modify force. (1 each)

(b) Mechanical advantage = Load ÷ Effort = 240 ÷ 60 = 4. Correct formula (1), answer (1).

(c) Any three explained, 2 marks each: multiply force; change force direction; increase speed or distance moved; reduce effort needed; enable lifting/moving loads more conveniently. (6)

23. Materials, joining and finishing (12 marks)

(a) Any four explained, 1 mark each: strength; durability; cost; availability; workability; weight; appearance; safety; environmental impact; suitability for intended use. (4)

(b) Any four methods with a correct description/use, 1 mark each: welding; soldering; riveting; screws/bolts; nails; adhesives; suitable wood joints. (4)

(c) Any two explained, 2 marks each: protect from corrosion/weather/wear; improve appearance; make surface easier to clean; increase service life. (4)

24. Energy and basic electrical systems (12 marks)

(a) Any three explained, 1 mark each: reduces dependence on fossil fuels; can lower long-term energy costs; reduces pollution; supports off-grid access; uses naturally replenished resources. (3)

(b) Award: complete closed-loop circuit (1); cell connected correctly (1); switch shown in series (1); bulb and wires labelled/connected correctly (1). Accept a correct open switch only if learner explains the circuit is then incomplete. (4)

(c) Any three: switch off supply before work; keep hands dry; avoid damaged wires; do not overload sockets; use correct fuse; keep water away; seek qualified help for mains wiring. (3)

(d) Any two: solar lighting; water heating; solar cooker; solar phone charging; solar irrigation pump. (1 each)

25. Workshop practice, safety and maintenance (12 marks)

(a) Any four rules with reasons, 1 mark each: wear PPE to prevent injury; keep floor clear to prevent trips; use tools correctly to avoid accidents; secure work to prevent slipping; follow teacher instructions; report hazards; avoid horseplay. (4)

(b) Any four steps, 1 mark each: identify task; select correct tool; inspect condition; wear suitable PPE; secure material; use correct technique; return tool safely after use. (4)

(c) Any four: clean work surfaces; return tools to labelled storage; remove waste safely; keep walkways clear; clean and dry tools; sharpen/lubricate as needed; report and repair damage. (4)

ADMINISTRATION NOTE

This is an original teacher-prepared assessment resource for practice and school-based assessment. Before administration, the Pre-Technical Studies teacher should confirm that the content and practical skills assessed match the actual Grade 9 Term 3 teaching sequence and the school scheme of work. For drawing questions, award marks for technically correct, clearly labelled alternatives.

Prepared by: Junior School Learning Hub